

=== GNB (1780742940) ===

```
cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.3
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "21401"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200

ru_sdr:
  device_driver: zmq
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6
  srate: 11.52
  tx_gain: 75
  rx_gain: 75

cell_cfg:
  dl_arfcn: 632000
  band: 78
  channel_bandwidth_MHz: 20
  common_scs: 15
  ssb_scs: 15
  plmn: "21401"
  tac: 7
  pdcch:
    common:
      ss0_index: 0
      coreset0_index: 6
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64

log:
  filename: /tmp/gnb.log
  all_level: info
  hex_max_size: 0

pcap:
  mac_enable: false
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: false
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: true
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap

metrics:
  layers:
    enable_rlc: true
    enable_sched: true
  periodicity:
    du_report_period: 1000
```

=== UE (1780742940) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 214010000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780742940) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
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#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
```

SUBSCRIBER_DB=214010000000001,00112233445566778899aabbccddeeff,opc,63BFA50EE6523365FF14C1F45F88737D,8000,9,10.45.1.2

```
-
  - NETWORK_NAME_FULL=srsRAN
  - NETWORK_NAME_SHORT=srsRAN
  - TZ=Europe/Madrid
privileged: true
ports:
  - "9898:9999/tcp"
# Uncomment port to use the 5gc from outside the docker network
# - "38412:38412/sctp"
# - "2152:2152/udp"
command: 5gc -c open5gs-5gc.yml
healthcheck:
  test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
  interval: 3s
  timeout: 1s
  retries: 60
networks:
  ran:
    ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
```

```
gnb:
  container_name: srsran_gnb
# Build info
image: srsran/gnb
build:
  context: ..
  dockerfile: docker/Dockerfile
```

```

args:
  OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's
the easiest to use if the 5gc is not in your PC
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

```

```

configs:
  gnb_config.yml:
    content: |
      cu_cp:
        amf:
          addr: 10.53.1.2
          port: 38412
          bind_addr: 10.53.1.3
          supported_tracking_areas:
            - tac: 7
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```

```

    pdcch:
      common:
        ss0_index: 0
        coreset0_index: 6
      dedicated:
        ss2_type: common
        dci_format_0_1_and_1_1: false
    prach:
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  ngap_enable: false
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: true
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap

metrics:
  layers:
    enable_rlc: true
    enable_sched: true
  periodicity:
    du_report_period: 1000
gnb_compose_config.yml:
  content: |
    cu_cp:
      amf:
        addr: ${OPEN5GS_IP:-10.53.1.2}
        bind_addr: 0.0.0.0
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1780742940) ===

1. PLMN (21401) sincronizado en gNB, UE (IMSI) y SUBSCRIBER_DB (Regla 1, 7a).
2. dl_arfcn (632000) seleccionado para Banda 78 (Regla 2).
3. common_scs (15) y ssb_scs (15) igualados en gNB (Regla 3).
4. Puertos ZMQ cruzados entre gNB y UE (Regla 4).
5. bind_addr configurado según Regla 5: gNB YAML con 10.53.1.3 y gnb_compose_config con 0.0.0.0.
6. Bloque ru_sdr copiado exactamente del CAG (Regla 6).
7. IMSI, K, OPC y formato de SUBSCRIBER_DB validados (Regla 7a, 7b).